Arpit Ramani

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Education

Georgia Institute of Technology — Atlanta, GA
B.S. in Computer Science (Threads: Theory & AI), GPA: 4.00 — *Expected May 2029*

- **Relevant Coursework:** Object-Oriented Programming, Multivariable Calculus, Linear Algebra, Discrete Mathematics

Wheeler Magnet School of Advanced Mathematics and Sciences — Atlanta, GA
GPA: 4.71, SAT: 1580 — *Sep 2022 – May 2026*

- **Awards:** Regeneron International Science and Engineering Fair 3rd Place Grand Award in Computational Biology and 2x Finalist; Society for Neuroscience and American Academy of Neurology presentations; Y Combinator Startup School 2026

Experience

Finaldose (Y Combinator P26) — London, UK
Machine Learning Engineer Intern — *Jun 2026 – Present*

- Trained a scikit-learn random forest to 0.852 AUROC over 1,368 SpCas9 positions under blocked cross-validation, beating the in-house baseline at 0.826 (paired-bootstrap 95% CI [+0.002, +0.046]) and ProDomino, the state of the art, at 0.777.
- Built the feature pipeline in Python/NumPy, pairing mean-field direct-coupling analysis over a 5,254-sequence alignment with coevolutionary graph-topology descriptors, then bounded the ceiling near 0.88 and traced the wall to label noise, not the model.
- Built a Bayesian Additive Regression Trees model with calibrated per-residue 95% credible intervals under domain-blocked cross-validation and a covariate-shift check abstaining on the 13% of sites outside measured support. (BART code)

Georgia Institute of Technology, Dr. Anqi Wu BrainML Lab — Atlanta, GA
Machine Learning Research Assistant — *Aug 2026 – Present*

- Built a training-time attribution method in PyTorch (model stitching, lockstep shadow models) for a 642k-parameter pose transformer, lifting correct stage attribution from 0.69 to 0.95 (95% CI +0.12 to +0.44) on a pre-registered planted benchmark.
- Showed that fine-tuning only the deepest stage raises attack F1 by 8.3% with no forgetting while full-model training on the same clips cuts it 7%, reproduced on a second machine within a 31-hypothesis pre-registered study.
- Built an FFT-basis Lasso surrogate (scikit-learn) searching 10 billion training curricula on AWS GPUs, beating random search 1.9x on held-out data, and used Hessian-vector products to show Adam's memory drives order effects (cosine 0.54 vs 0.00).

Emory University, Dr. Jingjing Yang Lab — Atlanta, GA
Computational Genomics Research Intern — *Jun 2024 – Present*

- Ran a Bayesian variable-selection model with spike-and-slab priors over 1,647 multi-omic ROSMAP samples and 1,000+ HPC compute hours, recovering 26 Alzheimer's risk genes that three published cis-only methods missed.
- Built the end-to-end GIFT fine-mapping pipeline across distributed Slurm job arrays, integrating GWAS summary statistics from 111,326 cases and 677,663 controls. (GIFT pipeline code)
- **First author** on an abstract accepted to the American Society of Human Genetics (ASHG) 2026 Annual Meeting.

Emory University, Dr. Blaine Roberts Lab — Atlanta, GA
Computational Proteomics Research Intern — *May–Aug 2026, Jul–Dec 2025*

- Built a differential-expression pipeline in R over a sparse 34-patient ALS cohort, chaining TAMPOR batch-effect normalization, random-forest imputation, and empirical-Bayes moderated tests under FDR control to recover power. (ALS pipeline code)
- Deployed a distributed molecular dynamics pipeline on AWS ParallelCluster, fanning replicas across Slurm GPU arrays with OpenMM and CUDA under an AMBER ff19SB/OPC force field at a 4 fs timestep. (MD pipeline code)
- Mapped 1,038 ALS and 1,628 Alzheimer's CSF nitropeptide sites against the FragPipe search database in Python, filtering to 506 and 479 high-confidence residues for conservation analysis.

Projects

Eidola | *Next.js, FastAPI, Postgres/pgvector, Docker, Cloud Run, Stripe, Swift* — Mar 2026 – Present

- Launched a multi-tenant memory app that grew to 106 active users, on a Next.js and FastAPI stack shipped through Cloud Build CI/CD to Cloud Run with row-level-security isolation, Stripe billing, 149 REST API endpoints, 103 test suites, and a 20.5k-line Swift iOS/watchOS client.
- Engineered the retrieval-augmented generation (RAG) layer with an HNSW index over 3,072-dim half-precision embeddings in PostgreSQL/pgvector, per-user HDBSCAN topic clustering, and answers streamed token by token over server-sent events.
- Earned $25,000 in cloud compute credits from Anthropic, Google Cloud, and AWS through Y Combinator Startup School.

Ferromilab | *Python, AlphaFold, MODELLER, Rosetta, HADDOCK, CHARMM QM/MM* — May 2024 – May 2025

- Automated an end-to-end structure-prediction and docking pipeline (AlphaFold and MODELLER refinement, HADDOCK epitope mapping, 1,352 grafted Rosetta CDRs) for bispecific antibody design against the hepcidin–ferroportin interface.

Technical Skills

Languages: Python, TypeScript, JavaScript, Swift, R, SQL, Bash
ML & Data: PyTorch, transformers, scikit-learn, Bayesian inference, interpretability, Bayesian optimization, pandas, NumPy, RAG
Tools & DevOps: Docker, Kubernetes, GKE, AWS, Google Cloud Run/Build, CI/CD, Slurm, CUDA, Git, Linux
Web & Backend: Next.js, React, FastAPI, PostgreSQL, REST APIs, Stripe, Supabase